

PROJECT GUIDE

Islet

A Dynamic-Island-style notch for Windows 11

Music, Claude Code and Codex activity, notifications, battery and more,
in one small black pill at the top of your screen.

MADE BY	VERSION	PLATFORM	BUILT WITH
Gopinath	0.1.0	Windows 10 / 11	Electron 37

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github.com/goprocker/Islet

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SECTION 01

Why Islet exists

Phones and Macs keep live status in one place at the top of the screen. Windows spreads it across the taskbar, the tray, a corner of banners and hidden terminal windows. Islet pulls it into one pill.

One glance

See what is playing, what your AI coding agent is doing, and new messages without switching windows.

Act in place

Skip songs, seek, change volume, and Allow or Deny Claude Code permission requests from the notch.

Out of the way

Click-through except over the notch, hides in fullscreen apps, and goes to a single dot when idle.

Local and light

No account, no cloud, no tracking, and no unsigned helper exe. Works with Smart App Control on.

The real thing



Open notch: agent sessions on the left, Spotify in the middle, time and battery on the right, tools at the edge.

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SECTION 02

Every feature

Grouped by area. Everything below ships in this repository today.

Music

- Spotify, or any player using Windows media controls (YouTube in a browser, VLC, ...)
- Album art, title, artist; accent colour taken from the art
- Previous / play-pause / next
- Click the bar to seek; elapsed and total time
- Scroll over the music side to change Spotify's own volume
- Like the song (Spotify's Alt+Shift+B shortcut)
- Live wave from Spotify's audio only (24 FFT bands)
- Timed lyrics from lrclib.net under the seek bar
- Enlarged album view over a blurred backdrop
- Song-change peek: the notch opens briefly on a new song
- Optional Spotify Web API for like state and up next (Premium)

Claude Code and Codex

- Sessions from both agents in one column, newest first
- Status per session: working, needs you, done
- Last 3 actions (prompt, tools, edited files)
- Allow / Deny Claude Code permission requests from the notch
- Session chips to switch project; Codex chips have a blue dot
- Click a project to open its folder
- One-click hook install with backups; untick to remove
- Falls back to reading transcripts when hooks are off

Notifications

- Every Windows toast becomes a notch pop-up
- Real app logos (Store manifests, shortcuts, exe icons)
- Click a pop-up to open the sending app
- Hover a pop-up to keep it on screen
- Windows' own bottom-right banners turned off while running, restored on quit
- Do Not Disturb, with an optional daily schedule

Battery and system

- Battery level and charging state
- Charging: bolt pop, pulsing green edge glow and a streak on the bottom edge
- Charging / unplugged pop-ups and a one-time low-battery warning
- Mic, camera and screen-capture dots while in use
- Clock and date column
- Hides while a fullscreen app is in front

Info

- Weather by city (Open-Meteo, no key), C or F
- Next calendar event from a private ICS link; chip 15 min before
- Browser download progress and a finished pop-up
- Timer from the tray, shown as a chip

Tools and window

- Clipboard history: last 5 texts, memory only
- File shelf: drop files in, drag them out
- Hover to open, click elsewhere to close
- Ctrl+Alt+N: pill, open, hidden; Ctrl+Alt+M: next monitor
- Quiet state: plain black pill with one dot when nothing is happening
- Optional idle line, sizes compact / normal / large
- Settings window, tray menu, start with Windows, single instance

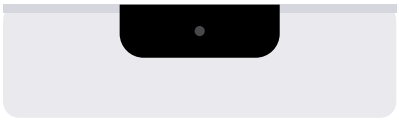
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SECTION 03

How it looks: every state

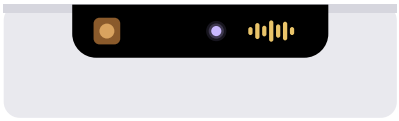
The notch is always black and hangs from the top edge of the screen. It changes size and content with what is going on. Drawings below are to scale with the real CSS sizes.

Closed pill



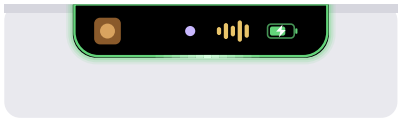
Quiet

Nothing playing, not charging, agent idle: one dot.



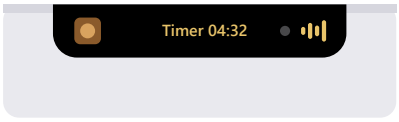
Playing

Album art, Claude dot (lit when active), live wave.



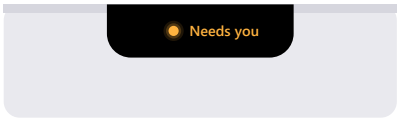
Charging

Green edge glow pulses; a light streak runs along the bottom.



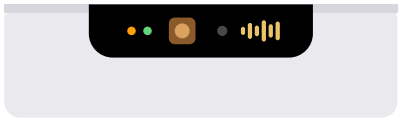
Timer / meeting / download

A chip shows the most urgent item.



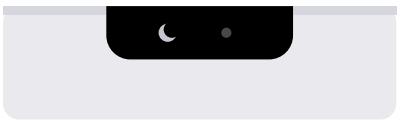
Agent needs you

Amber pulsing dot while a permission waits.



Mic / camera in use

Orange and green dots on the left.



Do Not Disturb

Moon icon; pop-ups are held back.

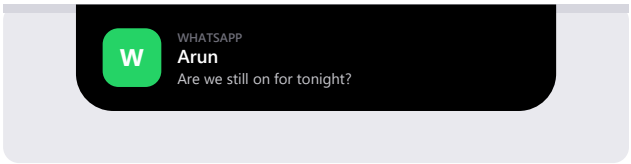
Idle line

Optional (idleHide): a thin 7 px line.

Hidden

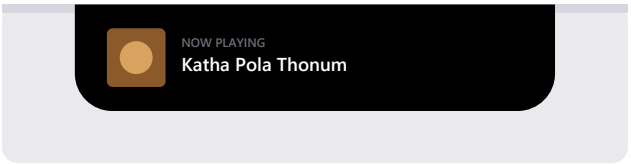
Ctrl+Alt+N or a fullscreen app: nothing on screen.

Pop-up and open



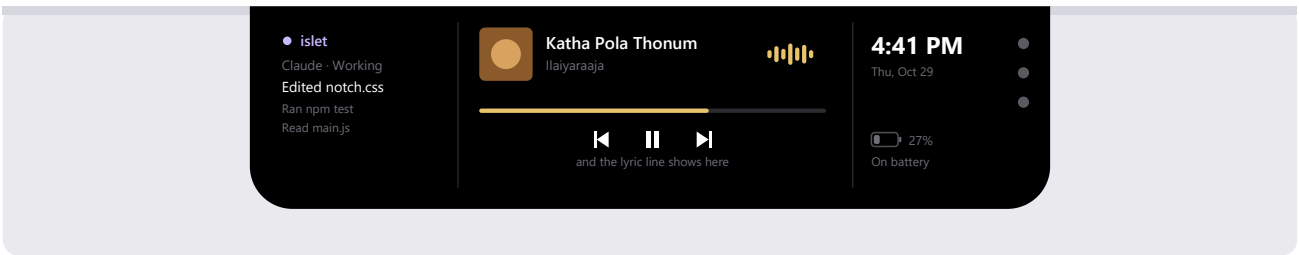
Notification pop-up

360 x 62 px with the real app logo. Click to open the app.



Song change peek

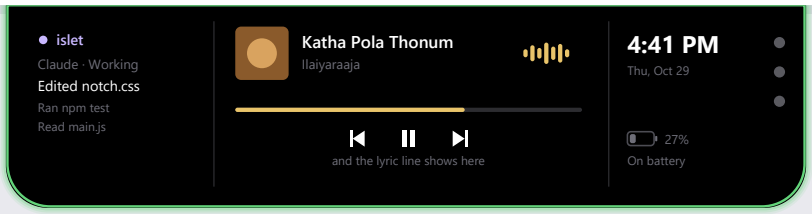
Opens briefly on a new song, then closes.



Open

Hover: 640 x 150 px. Agents, music, time and battery, tools. Click elsewhere to close.

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Open while charging

The same green glow follows the edge at every size.

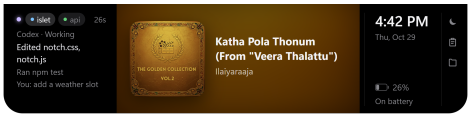
Real screenshots



Pill



Pop-up



Album view

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SECTION 04

Tech stack

Small and native to Windows. No UI framework, no database server, no cloud.

Layer	Technology	Used for
App shell	Electron 37 (Chromium + Node.js)	Frameless, transparent, always-on-top window; tray; global hotkeys; IPC
UI	HTML, CSS, vanilla JavaScript	The notch (renderer/) and the settings window (settings/)
Main process	Node.js 20+ (CommonJS)	Window control, config, timers, HTTP bridge, spawning helpers
Windows access	Windows PowerShell 5.1	Media session, privacy dots, fullscreen detection
Native code	C# compiled at runtime (Add-Type)	Spotify process-loopback audio capture, FFT, volume
Storage	node:sqlite (read-only), JSON	Notification database; %APPDATA%\islet\config.json
Registry	HKCU Notifications\Settings	Turning Windows' banners off and back on
Network	fetch()	lrclib.net, Open-Meteo, your ICS link, optional Spotify Web API
Agent hooks	HTTP server on 127.0.0.1:47321	Claude Code and Codex hooks post events
Tests	node --test	Bridge, calendar, downloads, hook install, stale responses
Packaging	electron-builder 25 (NSIS)	One-click per-user installer

Key decisions

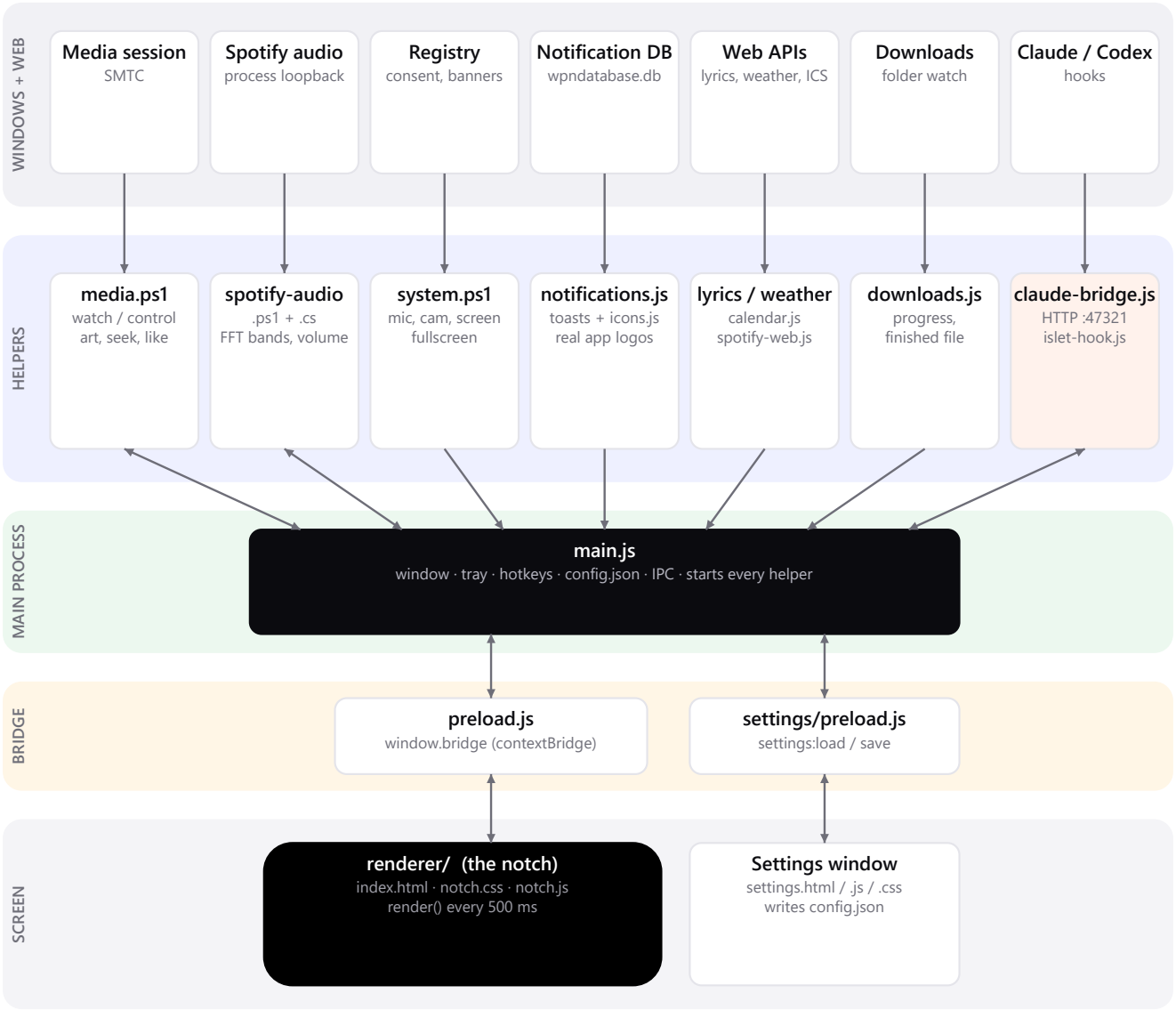
Decision	Why
No UI framework	One small view. Vanilla JS starts fast and is easy to read.
C# inside PowerShell	Smart App Control blocks unsigned exes. Add-Type compiles inside signed powershell.exe.
Read the notification DB	Windows has no notification-listener API for unpackaged apps.
Hooks over loopback HTTP	Instant agent status; transcript timing is the fallback.
Electron 37	Needed for node:sqlite.
Windows media controls	No Spotify login or Premium needed; works with other players too.

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SECTION 05

Architecture

Electron's two-process model. The main process talks to Windows and the web through helpers; the notch page only sees a small, safe API.



Double arrows are two-way: commands go back down (play, seek, volume, Allow / Deny).

How data moves

- **In:** each helper prints JSON lines (or POSTs, for hooks). main.js parses them and forwards with webContents.send.
- **Draw:** notch.js keeps the latest state and calls render() on every event and every 500 ms.
- **Out:** clicks and scrolls call window.bridge (ipcRenderer.send). main.js runs them: writes to media.ps1, spotify-audio stdin, or answers the waiting hook request.
- **Mouse:** the window is click-through. Entering the notch sends hover(true); main.js makes it clickable until the pointer leaves.

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SECTION 06

How each part works

Music	media.ps1 reads the Windows media session once a second (prefers Spotify while it plays) and prints JSON. Commands toggle, next, prev, seek and like come back on its stdin. spotify-audio captures only the Spotify process tree, turns it into 24 log-spaced bands about 30 times a second, and changes Spotify's own volume. lyrics.js looks up timed lyrics on lrclib.net and caches them per song.
Agents	claude-bridge.js runs an HTTP server on 127.0.0.1:47321 and keeps a status per session. hooks/islet-hook.js is what each hook runs: it POSTs the event. A PermissionRequest is held for permissionWaitSeconds so you can answer in the notch; with no answer the terminal asks as usual.
Notifications	notifications.js polls wpndatabase.db read-only every 2 s and parses the toast XML. icons.js finds a real logo. Banners are hidden with ShowBanner = 0 in the registry; the old values go to banners.json and come back on quit.
System	system.ps1 reads the CapabilityAccessManager consent store for mic, camera and screen use, and detects fullscreen windows (two equal readings before changing). Battery comes from the Battery API in the page.
Info	weather.js (Open-Meteo, every 15 min), calendar.js (next timed event in 24 h, simple repeats), downloads.js (.crdownload / .part / .partial files, stale ones ignored after 60 s).
Settings	settings/ is its own window with its own preload. main.js checks every value before saving config.json, watches the file, and applies hand edits within 2 s.

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SECTION 07

Repository map

Every tracked file, with its size in lines and what it is for.

Path	Lines	Purpose
main.js	605	Main process: window, tray, hotkeys, config, IPC, starts every helper
preload.js	43	window.bridge: the only API the notch page can call
renderer/index.html	101	Notch markup: pill, pop-up, open view, panels
renderer/notch.css	256	Notch styling, states and animations
renderer/notch.js	523	Open / close, music, agents, battery, pop-ups, shelf, render loop
settings/	158	Settings window: HTML, CSS, JS and its own preload
claude-bridge.js	177	Hook HTTP server; installs and removes Claude Code / Codex hooks
hooks/islet-hook.js	51	Run by each agent hook; POSTs the event
media.ps1	121	Windows media session reader and controller
spotify-audio.ps1 / .cs	325	Spotify loopback capture, FFT bands, volume
system.ps1	71	Mic / camera / screen dots, fullscreen detection
notifications.js	127	Notification database reader, banner toggle
icons.js	114	Real app logos
lyrics.js	46	lrclib.net timed lyrics
weather.js	47	Open-Meteo weather
calendar.js	81	Next event from an ICS link
downloads.js	53	Browser download progress
spotify-web.js	106	Optional Spotify Web API (PKCE login)
test/	225	5 test files run by node --test
demo.html	191	Original browser design demo (not used by the app)
docs/	-	Screenshots
package.json	53	Scripts, Electron version, installer config

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SECTION 08

Install and run

You need

- Windows 10 1903+ or Windows 11
- Node.js 20 or newer, with npm
- Git
- Windows PowerShell 5.1 (already part of Windows)

Run from source

```
git clone https://github.com/goprocker/Islet.git
cd Islet
npm install
npm start
```

The notch appears at the top of your main screen. On first run Islet adds itself to Start with Windows. Right-click the tray icon for the menu or open **Settings...**.

Build an installer (optional)

```
npm run dist      # creates dist\Islet Setup <version>.exe
```

The build is unsigned, so Smart App Control blocks it. Run from source, or sign it with your own certificate.

SECTION 09

Using Islet

Action	How
Open	Hover the notch
Close	Move away, or click anywhere else
Cycle pill / open / hidden	Ctrl+Alt+N
Next monitor	Ctrl+Alt+M, or tray > Show on screen
Volume	Scroll over the music side
Seek	Click the progress bar
Album view	Click the album art
Like	Heart button, or Alt+Shift+B
Clipboard / shelf / DND	Buttons in the tools column
Timer	Tray > Timer
Settings	Tray > Settings...

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SECTION 10

Claude Code and Codex

Tray > Connect to Claude Code and Codex installs hooks; untick it to remove them. Backups are saved next to each file.

Hook event	What the notch shows
UserPromptSubmit, PostToolUse	Working, with the latest actions
Notification, PermissionRequest	Needs you; Claude Code requests get Allow / Deny
Stop, SessionEnd	Done
SessionStart (Codex)	New session chip

- Claude Code: ~/.claude/settings.json (backup settings.json.islet-backup).
- Codex: ~/.codex/hooks.json if ~/.codex exists (backup hooks.json.islet-backup). Approve Islet's hooks once in Codex (/hooks).
- permissionWaitSeconds (default 8) is how long a request waits for you in the notch. 0 turns holding off.
- Codex permission requests show 'Needs you' but are answered in Codex.

SECTION 11

Settings

Stored in %APPDATA%\islet\config.json and edited from the settings window.

Key	Default	Meaning
size	normal	compact, normal or large
accent	album	Colour from album art, or #rrggbb
idleHide	false	Thin line when nothing is happening
hideInFullscreen	true	Hide behind fullscreen apps
notifications	true	Show Windows notifications in the notch
hideWindowsBanners	true	Turn off Windows' own banners while running
dnd / dndFrom / dndTo	off	Do Not Disturb, optional daily HH:MM window
permissionWaitSeconds	8	Allow / Deny hold time, 0 to 24
lyrics	true	Fetch lyrics from lrclib.net
weatherCity	empty	City name; empty turns weather off
weatherUnit	C	C or F
calendarIcsUrl	empty	Private iCal link
downloadsFolder	empty	Empty means Downloads
clipboardHistory	true	Keep the last 5 copied texts in memory
displayId	null	Screen to show on

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Key	Default	Meaning
spotifyClientId	empty	Optional Spotify Web API (Premium)

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SECTION 12

Build, test, debug

```
npm test        # node --test test/*.test.js
npm run dist    # NSIS installer

# inspect the notch with Chrome DevTools
node_modules\electron\dist\electron.exe . --remote-debugging-port=9333
```

Open chrome://inspect to see the page. Handy globals: setOpen(true), render().

SECTION 13

Privacy, troubleshooting, uninstall

Privacy

- Everything runs on your PC. Network calls only go to lrclib.net, Open-Meteo, your ICS link, and Spotify if connected.
- Clipboard history never touches disk. The shelf stores file paths only.
- The hook server listens on 127.0.0.1 only. The notch page has no direct Node access.
- Registry changes are limited to banner flags under HKCU and are put back on quit.

Troubleshooting

Problem	Fix
Banners still bottom-right	Start Islet normally (sandboxed shells can't write the registry). Sign out and in once if needed.
Banners off after a crash	Start Islet and quit from the tray, or turn the setting off.
Codex column empty	Approve Islet's hooks in Codex (/hooks).
Installer blocked	Smart App Control; run with npm start.
Second launch does nothing	Only one copy runs; it opens the settings of the running one.

Uninstall

- Tray: untick Connect to Claude Code and Codex, and Start with Windows.
- Tray: Quit (puts Windows banners back).
- Delete the folder and %APPDATA%\islet.

Islet

Designed and built by **Gopinath**

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